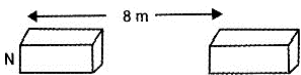


SOUND-05

Class 09 - Science

1. A battery lights a bulb. Describe the energy changes involved in the process. [2]
 2. A force of 7N acts on an object. The displacement is say 8 m, in the direction of the force. Let us take it that the force acts on the object through the displacement. What is the work done in this case? [2]
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3. Can any object have mechanical energy even if its momentum is zero? Explain. [2]
 4. A body of mass 4 kg initially at rest is subjected to a force of 16 N. What is the kinetic energy acquired by the body at the end of 10 s? [2]
 5. If an electric iron of 1200 W is used for 30 minutes every day, find the electric energy consumed in the month of April. [2]
 6. The work done in lifting a box on to a platform does not depend upon how fast it is lifted up. Explain your answer giving proper reason. [2]
 7. The volume of 50 g substance is 20 cm^3 . If the density of water is 1 g cm^{-3} , will the substance float or sink? [2]
 8. Calculate the work done against the gravity. [2]
 9. A horse pulls a cart with a force of 300 N, such that the system of horse and cart moves with uniform speed of 18 kmh^{-1} on a level road. Calculate the power developed by the horse in watt and also find its equivalent horse power. (Take $1 \text{ hp} = 746 \text{ W}$) [2]
 10. A boy of mass 40 kg runs up flight of 50 steps each 10 cm high in 5 second. Find [2]
 - i) the work done by the boy ii) the power developed. ($g = 9.8 \text{ ms}^{-2}$)
 11. A person is holding a bucket by applying a vertical force of 10 N. He moves a horizontal distance of 5 m and then climbs up a vertical distance of 10 m Calculate the total work done by him. [2]
 12. Why cutting instruments are sharpened? [2]
 13. If an electric bulb of 100 watt is lighted for 2 hours. how much electric energy would be consumed? [2]
 14. Calculate the work done in lifting 200 kg of a mass through a vertical distance of 6 m. Assume $g = 10 \text{ m/s}^2$. [2]
 15. An object of mass, m is moving with a constant velocity, v . How much work should be done on the object in order to bring the object to rest? [2]
 16. A 150 kg car engine develops 500 W for each kg. What force does it exert in moving the car at a speed of 20 ms^{-1} ? [2]
 17. Find the energy in joules consumed in 10 hours by four devices of power 500 W each. [2]
 18. A rocket is moving up with a velocity v . If the velocity of this rocket is suddenly tripled, what will be the ratio of two kinetic energies? [2]
 19. Suppose a hammer which falls freely on a nail placed on a piece of wood of mass 1 kg, if it falls from a height of [2]

- 1 m, how much kinetic energy it will have just before hitting the nail? Take $g = 10 \text{ ms}^{-2}$.
20. The Jog falls in Karnataka state are nearly 20 m high 2,000 tonnes of water falls from it in a minute. Calculate the equivalent power, if all this energy can be utilised. ($g = 10 \text{ ms}^{-2}$) [2]
 21. A mass of 10 kg falls from a height of 50 cm. Calculate (i) Velocity of the body (ii) Kinetic energy of body when it just reaches the ground level. Does the velocity depend upon the mass of the particle? Explain. (Take $g = 10 \text{ ms}^{-2}$) [2]
 22. Calculate the power of an engine required to lift 10^5 kg of coal per hour from a mine 360 m deep. [Take $g = 10 \text{ ms}^{-2}$] [2]
 23. Define average power. [2]
 24. A rocket of $3 \times 10^6 \text{ kg}$ mass takes off from a launching pad and acquires a vertical velocity of 1 kms^{-1} at an altitude of 25 km. Calculate [2]
 - a. Potential energy
 - b. Kinetic energy. (Take the value of $g = 10 \text{ ms}^{-2}$)
 25. Why does a block of plastic released under water come up to the surface of water? [2]
 26. The speed of a vehicle of mass 500 kg increases from 36 km/h to 72 km/h. Calculate the increase in its kinetic energy. [2]
 27. A person holds a bundle of hay over his head for 30 minutes and gets tired. Has he done some work or not? Justify your answer. [2]
 28. What is the work done by a coolie walking on a horizontal platform with a load on his head? [2]
 29. A mass of 10 kg is at a point A on a table. It is moved to a point B. If the line joining A and B is horizontal, what is the work done on the object by the gravitational force? Explain your answer. [2]
 30. Can any object have momentum even if its mechanical energy is zero? Explain. [2]
 31. Why a dead body floats, with its head immersed in water. [2]
 32. Compare the power at which each of the following is moving upwards against the force of gravity? (given $g = 10 \text{ ms}^{-2}$) [2]
 - i. a butterfly of mass 1.0 g that flies upward at a rate of 0.5 ms^{-1} .
 - ii. a 250-g squirrel climbing up on a tree at a rate of 0.5 ms^{-1} .
 33. An electric heater is rated 1500 W. How much energy does it use in 10 hours? [2]
 34. A certain household has consumed 250 units of energy during a month. How much energy is this in joules? [2]
 35. An object thrown at a certain angle to the ground moves in a curved path and falls back to the ground. The initial and final points of the path of the object lie on the same horizontal line. What is the work done by the force of gravity on the object? [2]
 36. Write an expression for the kinetic energy of an object. [2]
 37. What is the work to be done to increase the velocity of a car from 30 km h^{-1} to 60 km h^{-1} if the mass of the car is 1500 kg? [2]
 38. A light and a heavy object have the same momentum, find out the ratio of their kinetic energies. Which one has a larger kinetic energy? [2]
 39. The kinetic energy of an object of mass m moving with a velocity of 5 ms^{-1} is 25 J. What will be its kinetic energy when its velocity is doubled? What will be its kinetic energy when its velocity is increased to three [2]

times?

40. Two bodies of equal masses move with the uniform velocities v and $3v$ respectively. Find the ratio of their kinetic energies. [2]
41. Give any two uses of kinetic energy. [2]
42. An object of mass 40 kg is raised to a height of 5 m above the ground. What is its potential energy? If the object is allowed to fall, find its kinetic energy when it is half way down. [2]
43. A battery lights a bulb. Describe the energy changes involved in the process? [2]
44. Five bulbs each having 100 W power are used for 4 h, a heater having 1500 W power is used for 2 h and an electric iron of power 1000 W is used for 5 h. [2]
- i. Calculate the total energy consumed by them.
 - ii. Convert this energy into joules.
45. Avinash can run at a speed of 8 ms^{-1} against the frictional force of 10 N, and Kapil can move at a speed of 3 ms^{-1} against the frictional force of 25 N. Who is more powerful and why? [2]
46. Why it is easier for a man to swim in sea water than in river water? [2]
47. Explain why it is difficult to push a tin can into water keeping its mouth upwards than when its mouth is kept downwards towards the water? [2]
48. A balloon of volume 50 m^3 is filled with hot air of density 0.4 kgm^{-3} , if the weight of the fabric of balloon is 12 kgf and equipment 'P' is attached to it, such that balloon is in the state of equilibrium. Calculate: i) Weight of hot air, ii) Weight of hot air, balloon and equipment iii) Upthrust, iv) Weight of equipment. [2]
49. Two boys A and B weighing 60 kg and 40 kg respectively, climb on a staircase each carrying a load of 20 kg on their head. The staircase has 10 steps, each of height 50 cm. If A takes 20 s to climb and B takes 10 s to climb, then [2]
- i. who possesses greater power?
 - ii. find the ratio of their powers.
50. An engine can pump 30,000 litres of water a vertical height of 45 metres in 10 minutes. Calculate the work done by the machine and its power. [$g = 9.8 \text{ m/s}^2$; Density of water = 10^3 kg/m^3 , 1000 litre = 1 m^3]. [2]
51. The velocity of a body moving in a straight line is increased by applying a constant force F , for some distance in the direction of the motion. Prove that the increase in the kinetic energy of the body is equal to the work done by the force on the body. [2]
52. A certain household has consumed 250 units of energy during a month. How much energy is this in joule? [2]
53. Iceberg floats in sea. Why? [2]
54. Why will a sheet of paper fall slower than one that is crumpled into a ball? [2]
55. Two girls, each of weight 400 N climb up a rope through a height of 8 m. We name one of the girls A and the other B. Girl A takes 20 s while B takes 50 s to accomplish this task. What is the power expended by each girl? [2]
56. Give reason : Ice bergs floating in the sea are dangerous for the ship. [2]
57. What is power? How do you differentiate kilowatt from kilowatt-hour? The Jog Falls in Karnataka state are nearly 20 m high. 2000 tonnes of waterfalls from it in a minute. Calculate the equivalent power if all this energy can be utilized? ($g = 10 \text{ ms}^{-2}$) [2]
58. A man drops a 10 kg rock from the top of a 5 m ladder. What is its kinetic energy when reaches the ground? What is its speed just before it hits the ground? [2]
59. A machine raises a load of 750 N through a height of 16 m in 5 seconds. Calculate the power at which the [2]

machine works.

60. A force of 10 N displaces a body by a distance of 2 m at an angle 60° to its own direction. Find the amount of work done. [2]
61. What is the work done by the force of gravity on a satellite moving round the earth? Justify your answer. [2]
62. An object thrown at a certain angle to the ground moves in a curved path and falls back to the ground. The initial and the final points of the path of the object lie on the same horizontal line. What is the work done by the force of gravity on the object? [2]
63. A light body and a heavy body have the same kinetic energy. Which one will have the greater momentum? [2]
64. How are kinetic energy and momentum related? [2]
65. i. State and define SI unit of power. [2]
ii. A person carrying 10 bricks each of mass 2.5 kg on his head moves to a height 20 m in 50 s. Calculate power spent in carrying bricks of the person. (Given, $g = 10 \text{ ms}^{-2}$).
66. Which would have greater effect on kinetic energy of an object - doubling the mass, or doubling the velocity? [2]
67. A man whose mass is 50 kg climbs up 30 steps of a staircase, each measuring 20 cm high, in 30 s. Calculate the power used in climbing the stairs. (Take $g = 10 \text{ ms}^{-2}$) [2]
68. If a solid of the same density as that of a liquid is placed in it, what will happen to the solid? [2]
69. It is possible that an object is in the state of accelerated motion due to external force acting on it, but no work is being done by the force? Explain with example. [2]
70. Write an expression for the work done when a force is acting on an object in the direction of its displacement. [2]
71. Calculate the work required to be done to stop a car of 1500 kg moving at a velocity of 60 kmh^{-1} . [2]
72. Mountains roads rarely go straight up but wind up gradually. Why? [2]
73. Explain briefly why a balloon filled with helium gas rises in air? Why does the balloon not rise indefinitely? [2]
74. A force of 20 N displaces a body through a distance of 1 m at an angle of 60° from its own direction. Calculate the amount of work done. [2]
75. A girl having a mass of 35 kg sits on a trolley of mass 5 kg. The trolley is given an initial velocity of 4 ms^{-1} by applying a force. The trolley comes to rest after traversing a distance of 16 m. [2]
a. How much work is done on the trolley?
b. How much work is done by the girl?